Hariharan (Hari) Devarajan

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SUMMARY

- Computer scientist at LLNL specializing in data movement, storage systems, and performance analysis for large-scale scientific and AI workflows in HPC.
- Builds explainable I/O, performance monitoring, provenance, and workload-aware middleware systems that let scientists understand and optimize data movement across heterogeneous compute, storage, and software stacks.
- Creator of DLIO Benchmark (part of MLPerf Storage) and DFTracer; maintainer of GOTCHA. Published in SC, HPDC, IPDPS, ICS, CCGrid, Cluster, and TPDS with two best paper awards.

RESEARCH AREAS

- **Explainable I/O.** Explaining *why* I/O is slow by connecting low-level storage telemetry with workflow context and the anatomy of the storage stack: DataCrumbs (eBPF-based multi-layer I/O telemetry), Metis (agentic knowledge synthesis for explainable I/O, REX-IO'26), and the XIO vision (SSDBM'25).
- **Performance Monitoring and Analysis.** Scalable tracing and automated analysis of workflow data movement: DFTracer (SC'24), DFAnalyzer, WisIO (ICS'25), HORATIO (SSDBM'26), IOMax (SC-W'23), Recorder, and SmartIO (PDSW'25).
- **Provenance for Scientific Workflows (new).** Moirai (LDRD, PI), a provenance tool that tracks the lineage and lifecycle of data across workflow tasks, software layers, and storage systems.
- **Deep Learning I/O Characterization and Optimization.** From monitoring to characterization to optimization of AI training pipelines: DLIO (CCGrid'21, Best Paper; MLPerf Storage), GLANCED-IO (HPDC'26), CoDL (REX-IO'26), Stimulus (CCGrid'22), DYAD (SBAC-PAD'24), and checkpointing for large language models.
- **Workload-aware Middleware and Storage Libraries.** Hermes (HPDC'18), UnifyFS (IPDPS'26; R&D 100), DYAD (SBAC-PAD'24), WADO (SSDBM'26), HCL (Cluster'20), LABIOS (HPDC'19, Best Paper), and intent-driven I/O (Mimir, H5Intent, IFlux).

EXPERIENCE

Lawrence Livermore National Laboratory — Livermore, CA, USA
Computer Scientist — Sep 2023 – Current

- **Projects:** Moirai (PI, provenance), DLDL (Co-PI, deep learning data loading), Heuretes (data management thrust lead), XIO (I/O explainability), GOTCHA; past: FRACTALE (data management thrust lead), CODA (WADO, DPUs).
- PI of Moirai (LDRD ER), a new provenance tool tracking data lineage across workflow tasks and storage systems.
- Co-PI of DLDL (deep learning data loading); Data Management Thrust Lead for Heuretes and, previously, FRACTALE (built IFlux, intent-aware storage tiers and scheduling in Flux).
- Lead explainable I/O research (XIO): eBPF-based multi-layer telemetry (DataCrumbs) and agentic I/O explanation (Metis).
- Lead performance monitoring and analysis tools for AI and scientific workflows (DFTracer, DFAnalyzer, WisIO, HORATIO).
- Characterized and optimized the Rabbit near-node storage system for simulations and AI training (LLNL Director's Achievement Award).
- Built WADO, a distributed WORM storage service for asynchronous data operations (CODA); mentor four summer interns and manage two university subcontracts.

Lawrence Livermore National Laboratory — Livermore, CA, USA
Post Doctoral Researcher — June 2021 – Sep 2023

- Invented I/O intents for workload-aware storage (Mimir, H5Intent, TailorFS) under a DOE ASCR Early Career Award (IOPP).
- Worked on DYAD locality-aware data management, asynchronous checkpointing in SCR, and maintained the GOTCHA toolkit.

Illinois Institute of Technology — Chicago, IL, USA
Research Assistant — Aug 2016 – May 2021

- Designed multi-tiered storage systems and middleware: Hermes, HFetch, HCompress, HReplica, LABIOS, and HCL.
- Created DLIO, a data-centric benchmark for scientific deep learning (CCGrid'21 Best Paper).

Argonne National Laboratory — Lemont, IL, USA
Research Aid — May 2020 – Aug 2020

- Characterized I/O of AI applications at ALCF; built VaniDL combining Darshan traces with TensorBoard.

The HDF Group — Champaign, IL, USA
Research Intern — May 2018 – Aug 2018

- Designed a hierarchical buffering platform as an HDF5 VOL plugin, tested on the Cori supercomputer.

Oracle Financial Services Software — Mumbai, India
Application Developer 2 — Sept 2013 – July 2016

- Designed a REST API framework for product middleware; received the "We Applaud Award."

EDUCATION

Illinois Institute of Technology — Chicago, IL, USA
Ph.D. in Computer Science. CGPA: 3.66/4 — 2016–2021

National Institute of Technology — Jamshedpur, India
B.Tech (Hons) in Computer Science and Engineering. CGPA: 8.61/10 — 2009–2013

SELECTED PUBLICATIONS

- **Hariharan Devarajan**, Sarah Neuwirth, Chen Wang, Ray A. O. Sinurat, Neeraj Rajesh, Jay Lofstead, Vanessa Sochat, Daniel Milroy, and Thomas Scogland. "DataCrumbs: Efficient Cross-Layer Data Capture for Explainable I/O Performance in HPC Storage Stacks". *Proceedings of the SC'26 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis: International Workshop on Programming and Performance Visualization Tools (ProTools'26)*. 2026.
- Ray A. O. Sinurat, William Nixon, Philip Carns, Huihuo Zheng, Sandeep Madireddy, Sam Foreman, Troy Arcomano, Robert B. Ross, Haryadi S. Gunawi, and **Hariharan Devarajan**. "GLANCED-IO: Taming I/O Optimization for Deep Learning at Scale". *Proceedings of the 35th International Symposium on High-Performance Parallel and Distributed Computing (HPDC'26)*. 2026.
- **Hariharan Devarajan**, Brian Behlendorf, Blake Devcich, Dean Roehrich, Matt Richerson, Tony Floeder, Brent Petit, Don Boyd, Trace Hamada, Tim McCree, Gian-Carlo Defazio, James Corbett, Elsa Gonsiorowski, Cameron Stanavige, Nikoli Dryden, Roger Pearce, Kathryn Mohror, and Bronis R. de Supinski. "Chasing the Rabbit: A Systematic Exploration of Software-Defined HPC Storage". *Proceedings of the 7th Workshop on Extreme-Scale Storage and Analysis (ESSA'26), held in conjunction with the 40th IEEE International Parallel and Distributed Processing Symposium (IPDPS'26)*. 2026.
- **Hariharan Devarajan**, Vanessa Sochat, Daniel J. Milroy, and Tom Scogland. "IFlux: Intent-Aware Storage Tiers & Software Scheduling for HPC Systems". *IEEE Transactions on Parallel and Distributed Systems* (2026).
- Ian Lumsden, **Hariharan Devarajan**, Izzet Yildirim, Stefano Markidis, Andong Hu, Ivy Bo Peng, Luca Pennati, Dewi Yokelson, Stephanie Brink, Olga Pearce, Tom Scogland, Bronis R. de Supinski, Gian Luca Delzanno, Anthony Kougkas, Xian-He Sun, and Michela Taufer. "Optimizing I/O for an Exascale Implicit Kinetic Plasma Simulation using the Rabbit Storage System". *5th Workshop on Re-envisioning Extreme-Scale I/O for Emerging Hybrid HPC Workloads (REX-IO'25), 2025 IEEE International Conference on Cluster Computing Workshops (CLUSTER Workshops)*. 2025. **LLNL Director's Institutional Award**.
- Izzet Yildirim, **Hariharan Devarajan**, Anthony Kougkas, Xian-He Sun, and Kathryn Mohror. "WisIO: Automated I/O Bottleneck Detection with Multi-Perspective Views for HPC Workflows". *Proceedings of the 39th ACM International Conference on Supercomputing (ICS'25)*. 2025.
- **Hariharan Devarajan**, Loic Pottier, Kaushik Velusamy, Huihuo Zheng, Izzet Yildirim, Olga Kogiou, Weikuan Yu, Anthony Kougkas, Xian-He Sun, Jae Seung Yeom, and Kathryn Mohror. "DFTracer: An Analysis-Friendly Data Flow Tracer for AI-Driven Workflows". *SC24: International Conference for High Performance Computing, Networking, Storage and Analysis*. 2024.
- **Hariharan Devarajan**, Gerd Heber, and Kathryn Mohror. "H5Intent: Autotuning HDF5 with User Intent". *IEEE Transactions on Parallel and Distributed Systems* (2024).
- **Hariharan Devarajan**, Huihuo Zheng, Anthony Kougkas, Xian-He Sun, and Venkatram Vishwanath. "DLIO: A Data-Centric Benchmark for Scientific Deep Learning Applications". *2021 21st IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing (CCGRID)*. 2021. **Best Paper**.
- Anthony Kougkas, **Hariharan Devarajan**, Jay Lofstead, and Xian-He Sun. "LABIOS: A Distributed Label-Based I/O System". *Proceedings of the ACM 28th International Symposium on High-Performance Parallel and Distributed Computing (HPDC'19)*. 2019. **Best Paper**.
- Anthony Kougkas, **Hariharan Devarajan**, and Xian-He Sun. "Hermes: A Heterogeneous-Aware Multi-Tiered Distributed I/O Buffering System". *Proceedings of the ACM 27th International Symposium on High-Performance Parallel and Distributed Computing (HPDC'18)*. 2018.

Full list of 45+ publications in the CV. Google Scholar: 432 citations, h-index 12.

OPEN-SOURCE SOFTWARE

- **DLIO Benchmark** (https://github.com/argonne-lcf/dlio_benchmark) – used for Aurora acceptance testing at ALCF; part of MLCommons MLPerf Storage.
- **DFTracer** (https://github.com/llnl/dftracer) – multi-layer data-flow tracing; used by MLPerf Storage, ALCF, and OLCF.
- **GOTCHA** – lead developer and maintainer of LLNL's function-interposition library for DOE/NNSA tools.

HONORS AND SERVICE

- **Awards:** LLNL Director's Institutional Award (iPIC3D I/O on Rabbit, REX-IO'25); R&D 100 Award (UnifyFS); Best Paper, CCGrid'21; Karsten Schwan Best Paper, HPDC'19; Outstanding Service Award, Cluster'21.
- **Leadership:** Program Co-Chair, SSDBM 2026; Posters Co-Chair, HPDC 2026 and Cluster 2026; Keynote, REX-IO'26; Panelist, PDSW'25 and HPDC 2026.
- **Program committees:** IPDPS, HPDC, Cluster, CCGrid, ICS, ICPP, PDSW, ESSA, REX-IO; Review Board, IEEE TPDS; 200+ reviews.
- **Judging:** PACT ACM Student Research Competition; IPDPS PhD Forum (2024, 2026). **Funding:** PI, LDRD ER 27-ERD-025 ($750K); data management lead, LDRD SI (Heuretes).